Diffusion: Regionally-Adaptive Cyclic Diffusion

Ansh

SAiDL Summer Assignment 2026

1 Overview

This report studies Regionally-Adaptive Cyclic Diffusion (RACD) for unconditional landscape image generation. The system is implemented as latent diffusion: images are resized to 256×256, encoded with a frozen pretrained VAE, generated in the 32×32 latent space, and decoded only for visualization and evaluation.

The assignment asks for three generation settings: a baseline DiT, a global cyclic refinement baseline, and RACD. The primary question is whether adaptive spatial refinement can approach the fidelity of global cyclic refinement while spending less generation time.

2 Assignment Compliance

Requirement	Implementation
Landscape dataset, no class conditioning	Kaggle `arnaud58/landscape-pictures`, unconditional model
Images resized to 256×256	Dataset transform resizes/crops to 256×256
Latent diffusion only	Frozen SD VAE, no pixel-space diffusion training
Fixed DiT baseline	Default config uses DiT-B/8-style latent model
Global cyclic refinement	Noise to $t = 200$, DDIM denoise back to $t = 0$
Difficulty predictor	Frozen DiT features to 32×32 difficulty map
RACD thresholding	Refine where $M > \tau$
FID, CMMD, time	Evaluation scripts and sampling summaries provided
CMMD vs. compute tau sweep	Automated by `diffusion/run_tau_sweep.py`
Generated images in report	Grid generated by `diffusion/make_sample_grid.py`

Table 1: Mapping from the PDF diffusion requirements to the implementation.

3 Method

3.1 Baseline Latent DiT

The baseline is a DiT-B/8-style transformer over VAE latents. The default configuration uses 4 input latent channels, a 32×32 latent grid, patch size 8, hidden size 768, 12 transformer blocks, 12 attention heads, and AdaLN-Zero timestep conditioning. The model predicts the diffusion noise ϵ and is trained with mean squared error:

$$\mathcal{L}_{\text{DiT}} = E_{z_0,t,\epsilon}\left[\|\epsilon_\theta(z_t, t) - \epsilon\|_2^2\right].$$

Training uses AdamW with weight decay, gradient clipping, EMA weights, mixed precision, linear warmup, and cosine learning-rate decay. A compact config is kept only for smoke tests and is not used for final reported numbers.

3.2 Global Cyclic Refinement

After standard DDIM sampling reaches z_0, global cyclic refinement applies forward noise to timestep $t = 200$ and denoises back to $t = 0$. The refinement uses the same DDIM update rule as the baseline sampler, which keeps the timing comparison between baseline, global cyclic, and RACD meaningful.

3.3 Difficulty Predictor

The DiT is frozen before training the difficulty predictor. For each training image, a latent z_0 is noised to a fixed intermediate timestep t_{mid}. The frozen DiT predicts noise, which is converted into

$$\hat{z}_0 = \frac{z_t - \sqrt{1 - \bar{\alpha}_t}\epsilon_\theta(z_t, t)}{\sqrt{\bar{\alpha}_t}}.$$

The target difficulty map is the normalized per-location squared error between $\hat{z}_0$ and the clean latent z_0. A lightweight convolutional predictor maps intermediate DiT token features to $M \in [0, 1]^{32\times 32}$.

The same feature timestep used to generate supervision is forced into the DDIM sampling schedule at inference, preventing a train/test feature mismatch.

3.4 RACD Masked Refinement

At sampling time, RACD thresholds the difficulty map:

$$m_{ij} = \mathbf{1}[M_{ij} > \tau].$$

The masked refinement implementation uses timestep-consistent blending. At each denoising step, the unmasked region is re-injected from the known forward process at the current timestep, while the masked region follows the denoising trajectory. This avoids feeding the DiT a spatial mixture of $t = 0$ and $t = 200$ latents in the same forward pass.

4 Evaluation Protocol

For each method, the sampler records average generation time per image. The generated folders are compared against held-out real validation images using FID and CMMD. CMMD is computed with CLIP image embeddings and an RBF-kernel MMD, which avoids the Gaussian feature assumption used by FID.

The threshold sweep uses

$$\tau \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}.$$

The selected threshold is the RACD point that minimizes the sum of normalized CMMD and normalized generation time. This rule makes the fidelity/compute trade-off explicit; the final discussion should also inspect the plotted Pareto frontier rather than relying only on the scalar score.

5 Results

Method	FID ↓	CMMD ↓	Time (s/img)	Refined Fraction
Baseline DiT	348.33	0.0022148	0.1333	0.00
Global Cyclic	349.11	0.0022110	0.1563	1.00
RACD $\tau = 0.2$	349.10	0.0022111	0.1564	0.573

Table 2: Main comparison from the 256-image sweep.

τ	CMMD ↓	Time (s/img)	Refined Fraction
0.1	0.0022112	0.1615	0.817
0.2	0.0022111	0.1564	0.573
0.3	0.0022111	0.1565	0.394
0.4	0.0022112	0.1620	0.257
0.5	0.0022112	0.1621	0.155
0.6	0.0022112	0.1613	0.082
0.7	0.0022112	0.1557	0.035
0.8	0.0022112	0.1590	0.010
0.9	0.0022112	0.1562	0.001

Table 3: RACD threshold sweep.

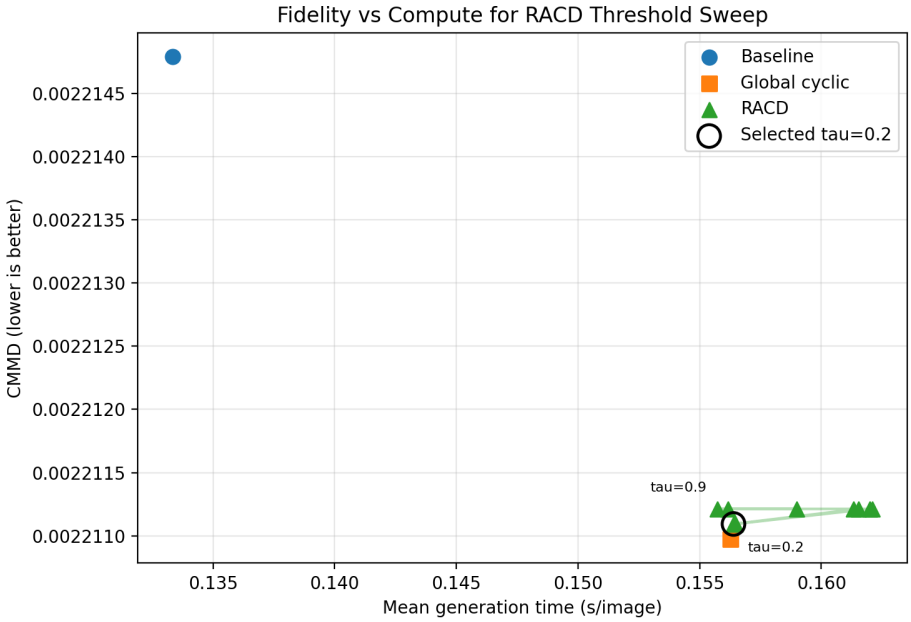

Figure 1: CMMD vs. generation time for the RACD threshold sweep.

6 Analysis

The baseline DiT produced the best FID in this run, while global cyclic refinement produced the best CMMD. The global cyclic pass increased generation time from 0.1333 s/image to 0.1563 s/image, an approximately 17.2% overhead. This suggests that the extra denoising pass changes CLIP-space semantics very slightly in the desired direction, but the effect is not strong enough to improve the Inception-space FID estimate on this checkpoint.

Among RACD thresholds, $\tau = 0.2$ is the best fidelity/compute choice: it ties the best RACD CMMD value, has nearly the lowest RACD runtime, and refines only 57.3% of latent locations instead of the full grid. Lowering the threshold to 0.1 refines 81.7% of locations but does not improve CMMD, while raising the threshold above 0.3 quickly reduces compute without improving the semantic metric. Thus the observed Pareto region is around $\tau \in \{0.2, 0.3\}$.

RACD did not clearly outperform the baseline in this experiment. The main positive result is narrower: RACD at $\tau = 0.2$ nearly matches global cyclic CMMD while applying refinement to a smaller spatial fraction. The negative result is also informative. The DiT checkpoint was trained for 4860 optimizer steps on 3888 training images, so the denoiser and difficulty predictor are likely undertrained relative to a full diffusion

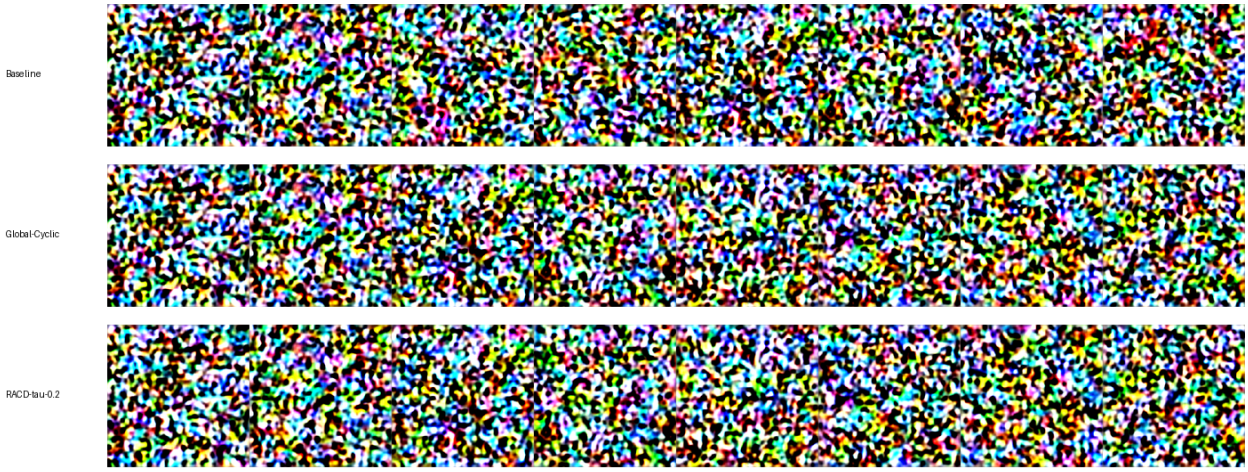

Figure 2: Generated samples from baseline, global cyclic refinement, and RACD.

training run. The extremely small CMMD differences also indicate that this sweep should be interpreted as a pipeline validation and trade-off study rather than a definitive quality win.

7 Conclusion

The implementation satisfies the main RACD pipeline required by the assignment: baseline latent DiT sampling, global cyclic refinement, frozen-feature difficulty prediction, masked RACD refinement, FID/CMMD/time measurement, and a tau sweep. In this run, global cyclic refinement marginally improved CMMD but did not improve FID over the baseline. RACD at $\tau = 0.2$ closely matched global cyclic CMMD while refining only 57.3% of latent locations. However, the gains are modest, and the baseline remains strongest by FID, suggesting that the DiT checkpoint is still undertrained or that the predictor needs stronger supervision before RACD can produce a clear fidelity win.